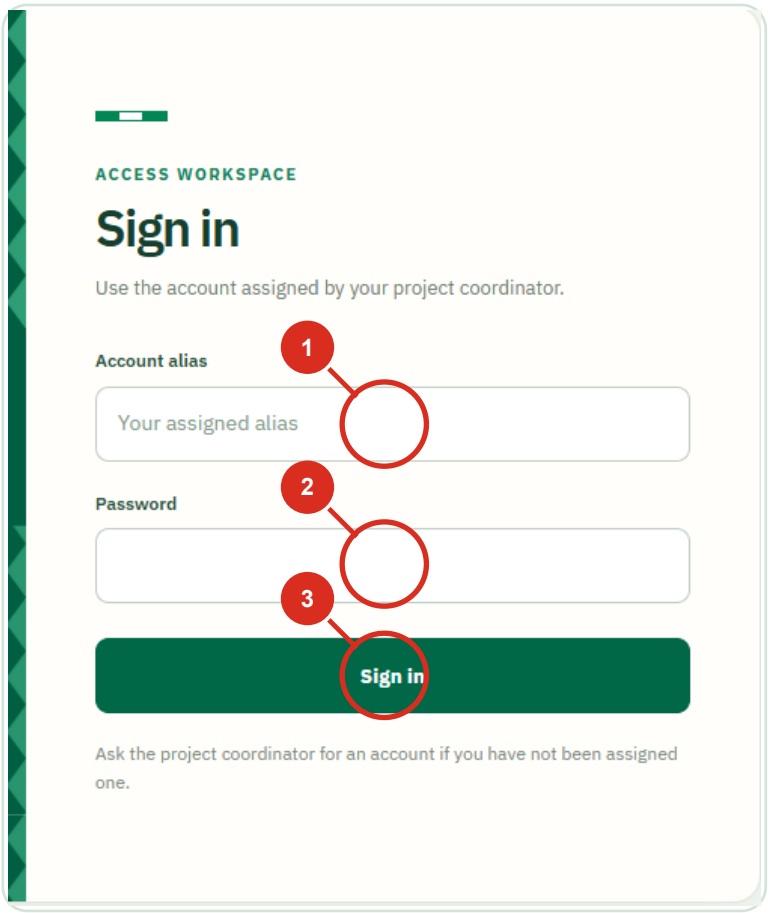
ResultTrail.

# User guide

## Observer and reviewer pathways

Capture a result copy / sync it / review the evidence

Use your assigned account at resulttrail.ng. This guide covers the two role pathways in the current working site. Screenshots show illustrative records and may differ from your assigned workspace.




1 Enter the assigned account alias. 2 Enter its password. 3 Select Sign in. Ask the project coordinator for credentials; they are intentionally absent from this guide.

THE PROJECT

# Why ResultTrail exists

An independent evidence workflow for illustrative polling-unit result copies.

What it is

ResultTrail is a web app for an observer team and its review desk. A field reporter records the figures from a polling-unit result copy. The app keeps an encrypted draft on that device, sends a saved version when the network is available, and shows a signed receipt when the service has received it. A reviewer can inspect the figures and source evidence, then record a decision with a traceable history.

Why I built it

I wanted to make a common handoff problem easier to manage: a team should be able to tell which copy was recorded, whether it really reached the shared service, and what a reviewer did with it. Weak connectivity can interrupt that handoff. A saved draft, a received version and an approved version should never be confused with one another.

What it is meant to solve

ResultTrail connects capture, delivery and human review in one workflow. It helps prevent an unsent report being mistaken for a delivered one, keeps corrections as new versions, surfaces basic arithmetic differences for source checking, and lets a reviewer produce a signed, minimized export after approval.

## THE WORKFLOW

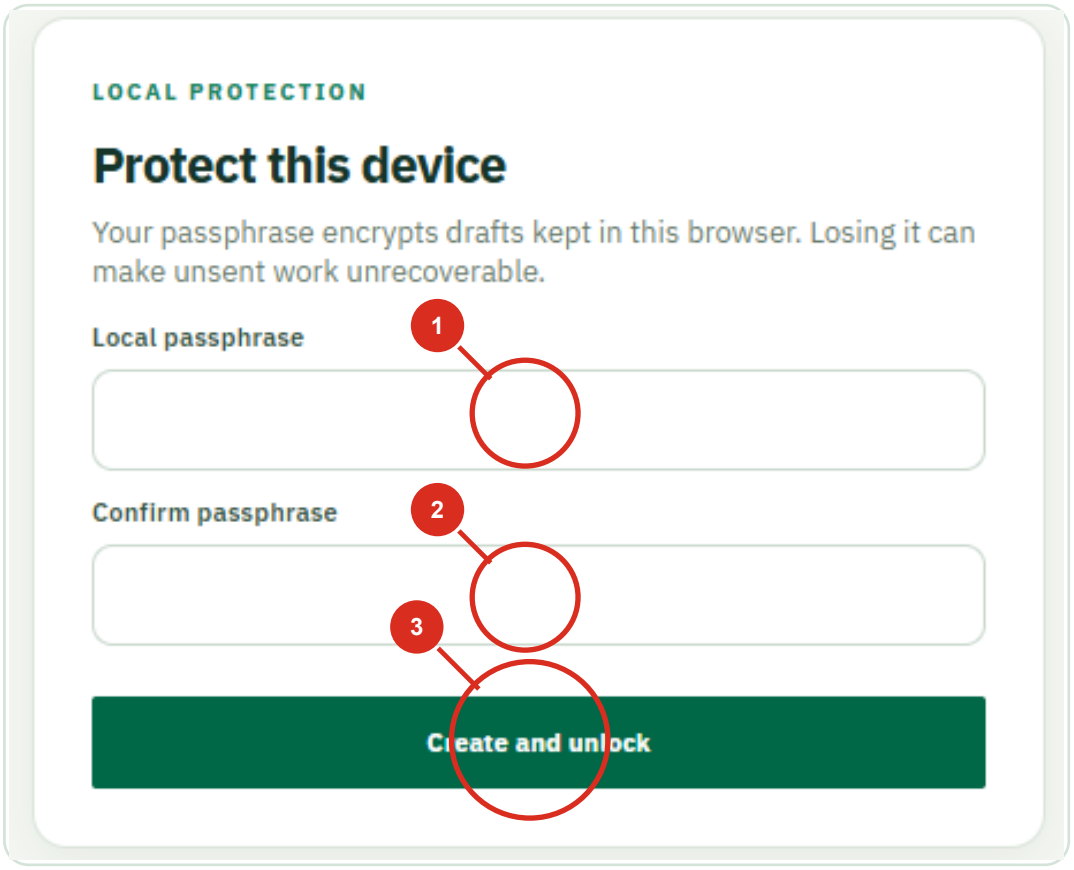
1. **CAPTURE** — Observer records the copy
2. **RECEIVE** — Service returns a signed receipt
3. **REVIEW** — Human reviewer records a decision

Scope of this guide. The screenshots show fictional demonstration records. ResultTrail is independent and does not publish or certify official election results. A receipt proves the service accepted a version; it does not prove that a paper copy or its figures were authentic.

OBSERVER PATHWAY

# 1. Protect this device

After the first observer sign-in, set a local passphrase for encrypted drafts in this browser.




### 1 Create a local passphrase

Enter a passphrase of at least 12 characters. It protects drafts saved on this device and is separate from your account password.

### 2 Confirm it

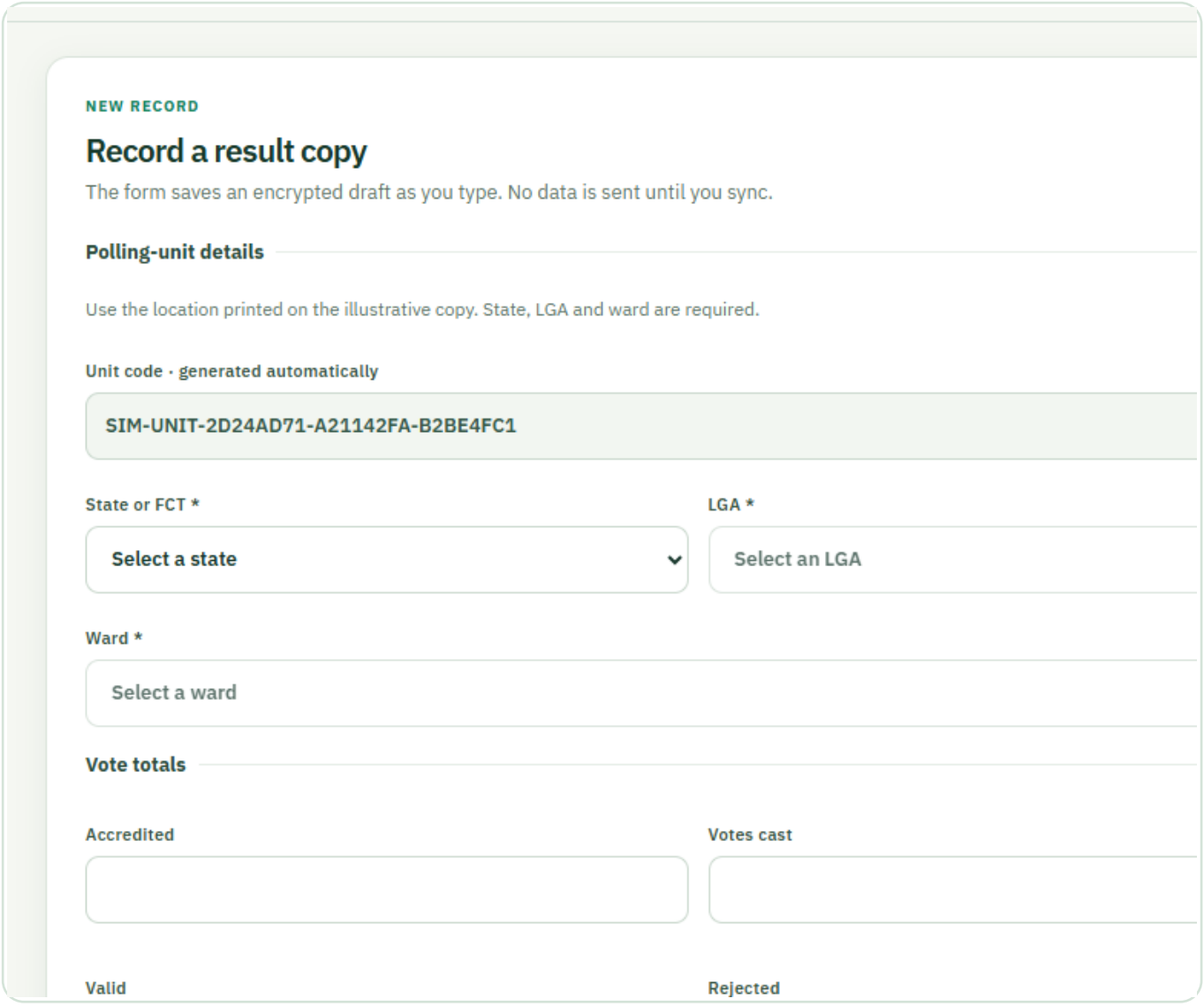
Re-enter the same passphrase, then choose Create and unlock (3). Keep it somewhere safe: losing it can make unsent drafts unrecoverable.

Returning later? Sign in, unlock the device vault, and resume your encrypted draft or saved queue. Lock device hides local drafts until you unlock again.

OBSERVER PATHWAY

# 2. Record the figures

Open New report. Copy the location and totals from the source result copy, not from memory.




1 **Identify the polling unit**

The site generates a unique read-only SIM-UNIT code for the new report. Select the state or FCT, LGA and ward shown on the illustrative copy; all three location fields are required.

2 **Enter the totals exactly**

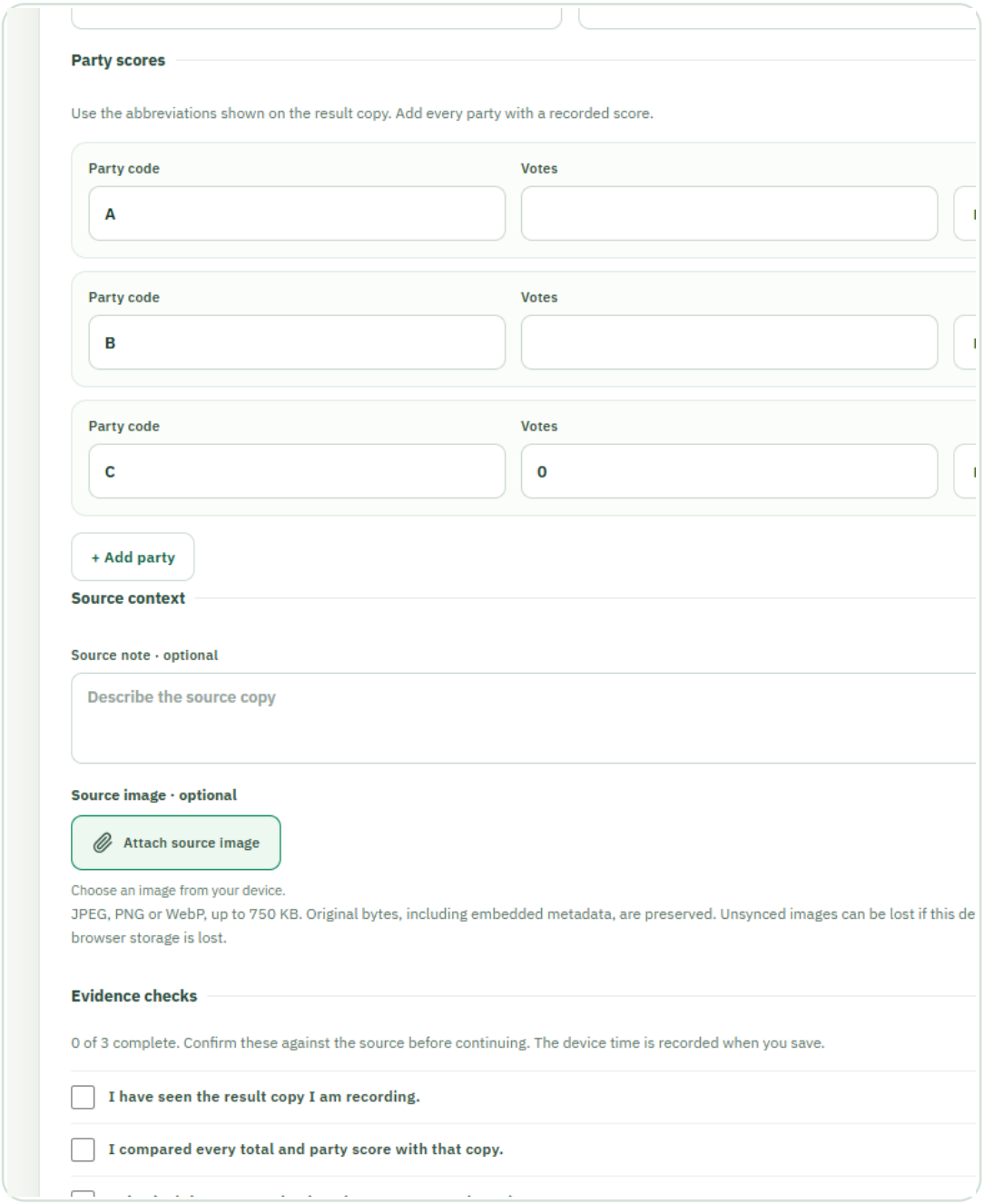
Record accredited voters, votes cast, valid votes and rejected votes as printed. The screenshot's numbers are a guide example only.

The draft saves encrypted on this device as you type. Nothing is delivered to the shared service until you sync a saved record.

OBSERVER PATHWAY

# 3. Add parties and check

Add every party shown on the copy; the form is not limited to three.




### 1 Add each party

Use + Add party for each additional party. Enter its code or abbreviation and its votes; use Remove only when the row does not belong.

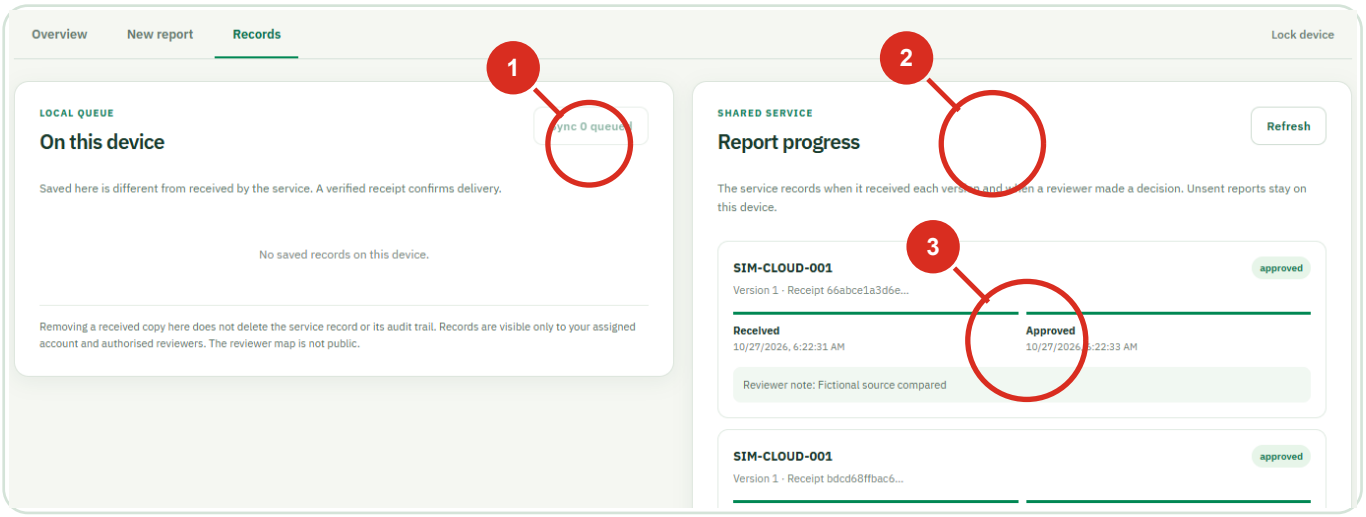
### 2 Check the source

Add a short source note and, if available, attach a JPEG, PNG or WebP source image up to 750 KB. The original is encrypted on this device. Tick the three checks after comparing the copy, figures and location.

Select Review figures, inspect the summary, then save the encrypted record. Required fields and capture checks must be complete before submission; an oversized or unsupported image shows an error.

# 4. Sync and follow progress

Open Records after saving. A local save is not yet a server receipt.




### 1 Sync a queued record

In On this device, choose Sync queued when connected. This example browser has no queued record, so its button is disabled.

### 2 Confirm delivery

Use Report progress to look for a Received entry and signed receipt. Refresh to check later reviewer decisions and notes. A local save alone does not prove delivery.

### 3 Verify the image

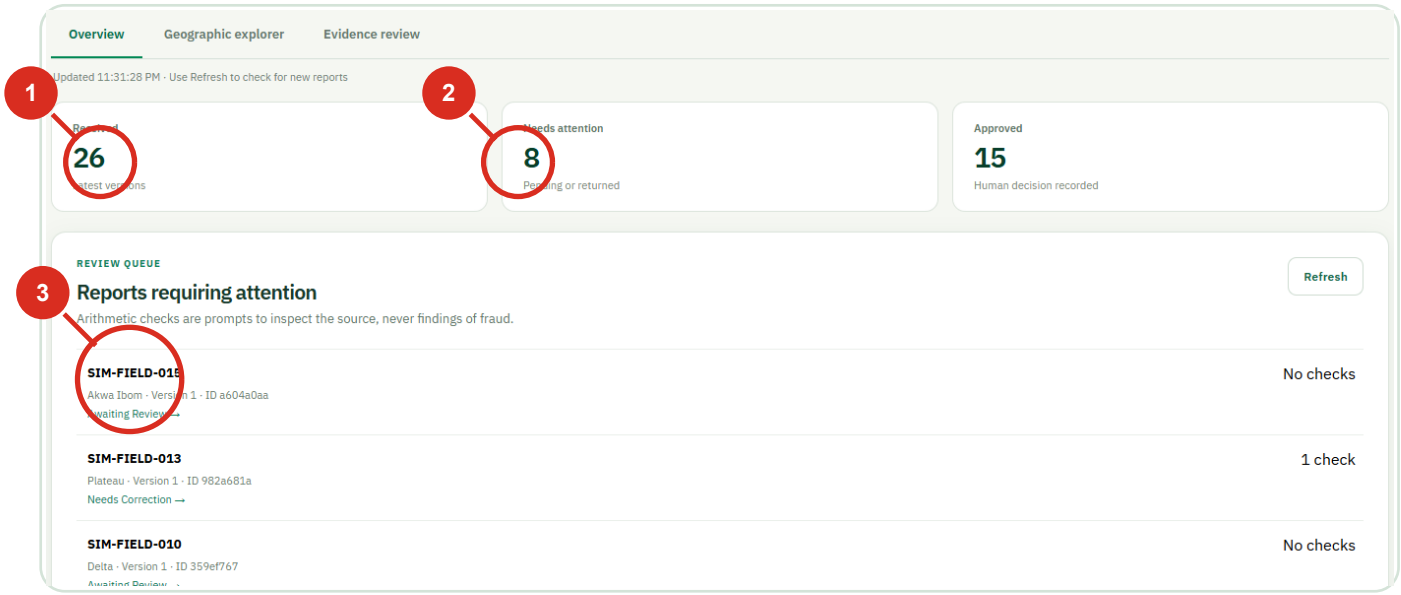
An image attached in the form stays encrypted with the unsent report. Sync sends the report first, verifies its signed receipt, then uploads the image. Records shows whether the image is pending or verified; retry Sync after a failed upload.

If a reviewer requests correction, compare the note with the source, submit a new version, and sync it. Keep the original evidence copy available for review.

REVIEWER PATHWAY

# 1. Find reports to review

Sign in with a reviewer account, then use Overview to see the queue and current counts.




### 1 Read the summary

Received counts the latest versions. Needs attention covers pending or returned versions; Approved counts reports with a recorded human decision.

### 2 Open a queue item

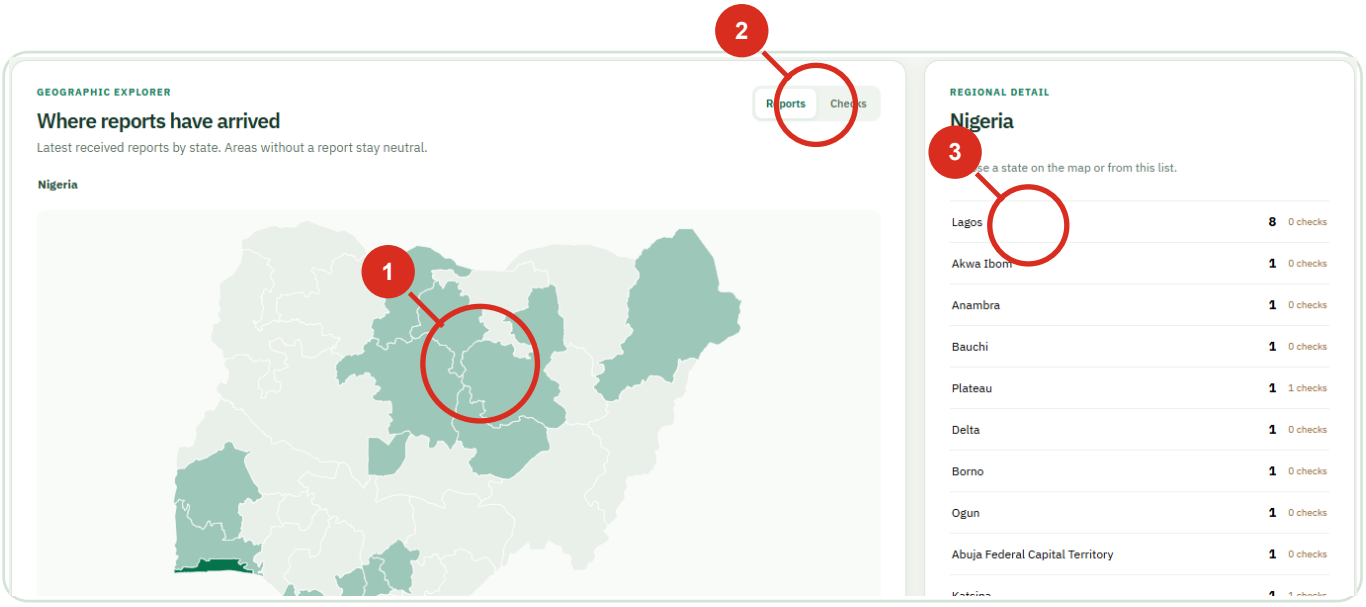
Select a report in Reports requiring attention, or open Evidence review and search by unit, state or LGA.

Arithmetic checks are prompts to inspect the source. They are not findings of fraud, and automated anomaly ranking is not active in this workspace.

REVIEWER PATHWAY

# 2. Explore the geography

The map helps locate submitted reports; it does not display election outcomes.




### 1 Start at Nigeria

Select a state on the map or in the regional list. Shading reflects where reports have arrived; unshaded areas remain neutral.

### 2 Drill down

Within a state, select an LGA; within an LGA, select a ward where available. The map zooms and the regional detail updates at each level.

### 3 Switch the layer

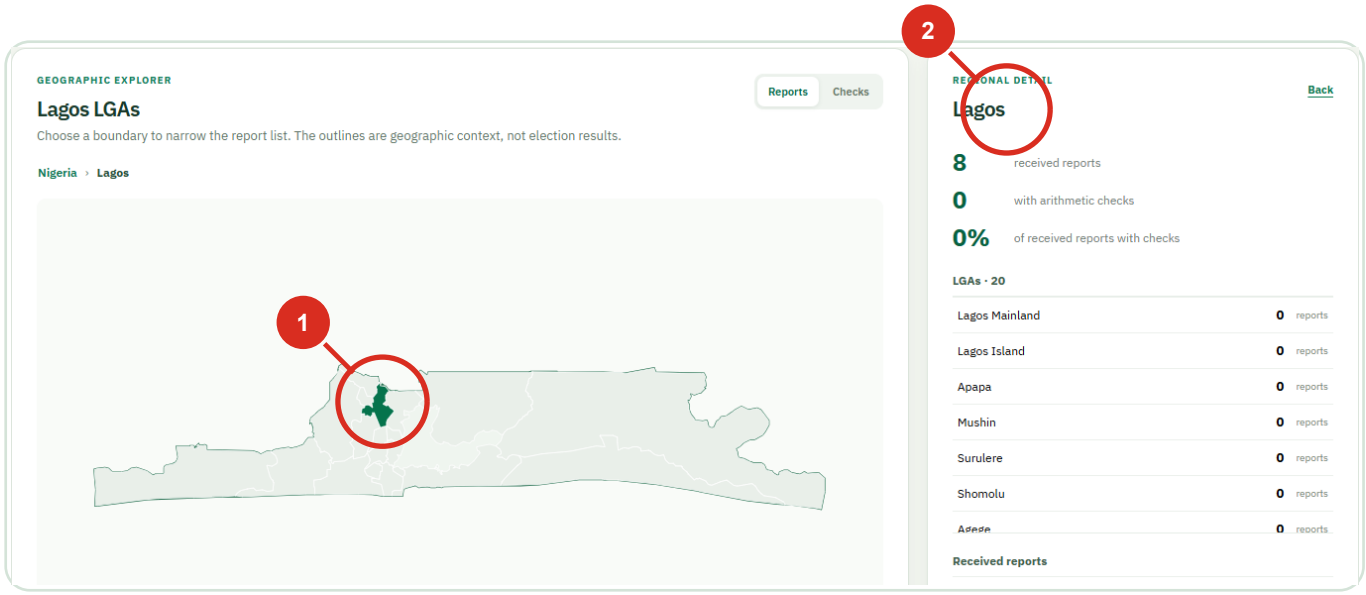
Use Reports for submission coverage or Checks for counts of arithmetic checks. The checks layer is a review aid, not a verdict.

Example path in this guide: Nigeria → Lagos → Ikeja → Aguda. Select a report from the region to open its evidence review.

REVIEWER PATHWAY

# 3. Open a state

Example: select Lagos from the Nigeria map or the state list.




### 1 See Lagos LGA borders

The state view outlines Lagos's local government areas. Select the shaded Ikeja area on the map, or choose Ikeja from the LGA list in Regional detail.

### 2 Read the regional summary

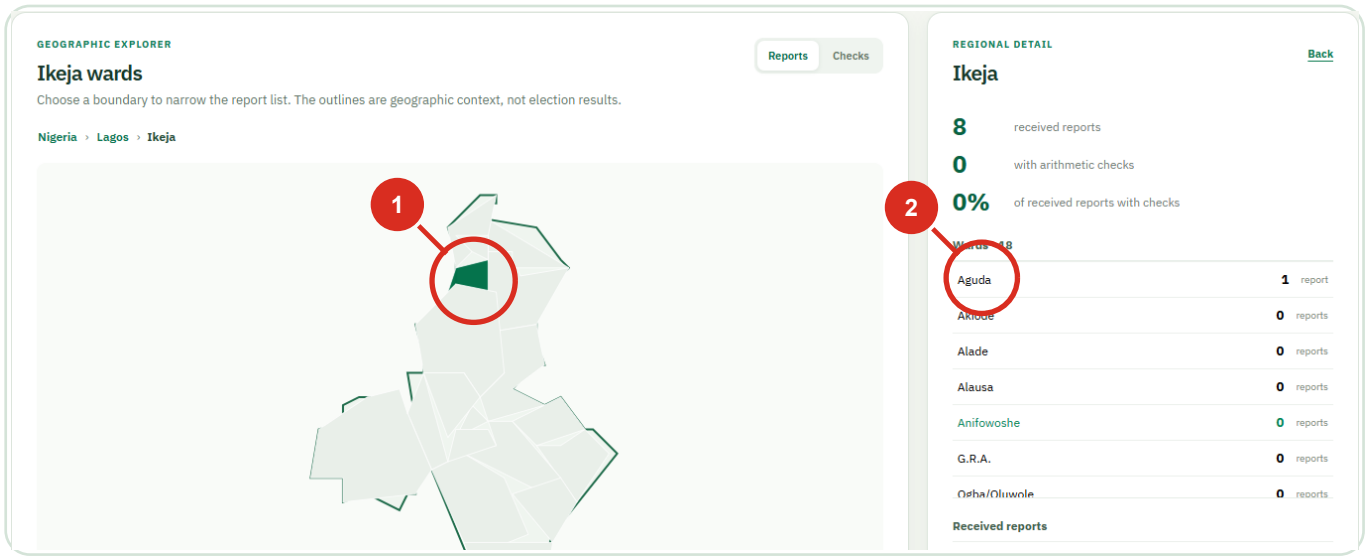
The panel shows received-report and arithmetic-check counts for the selected state. The map outlines are geographic context, not vote totals.

Use the breadcrumb above the map or Back in Regional detail to return to Nigeria.

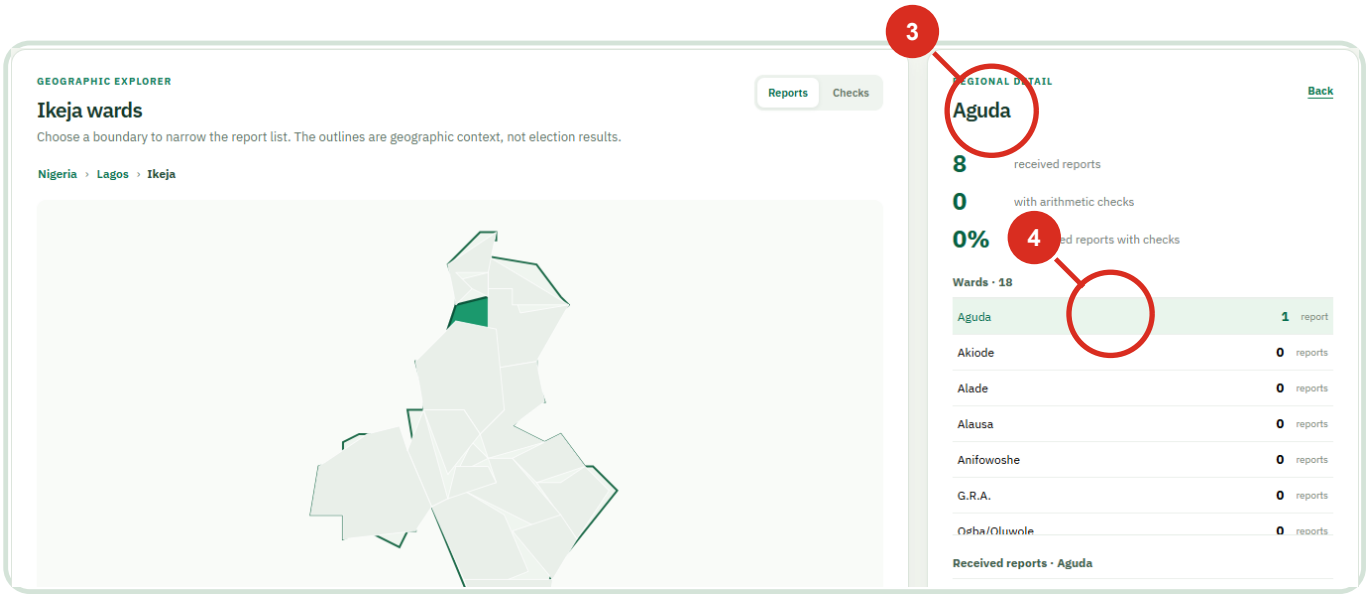
REVIEWER PATHWAY

# 4. Open an LGA and ward

Example: Lagos → Ikeja → Aguda.



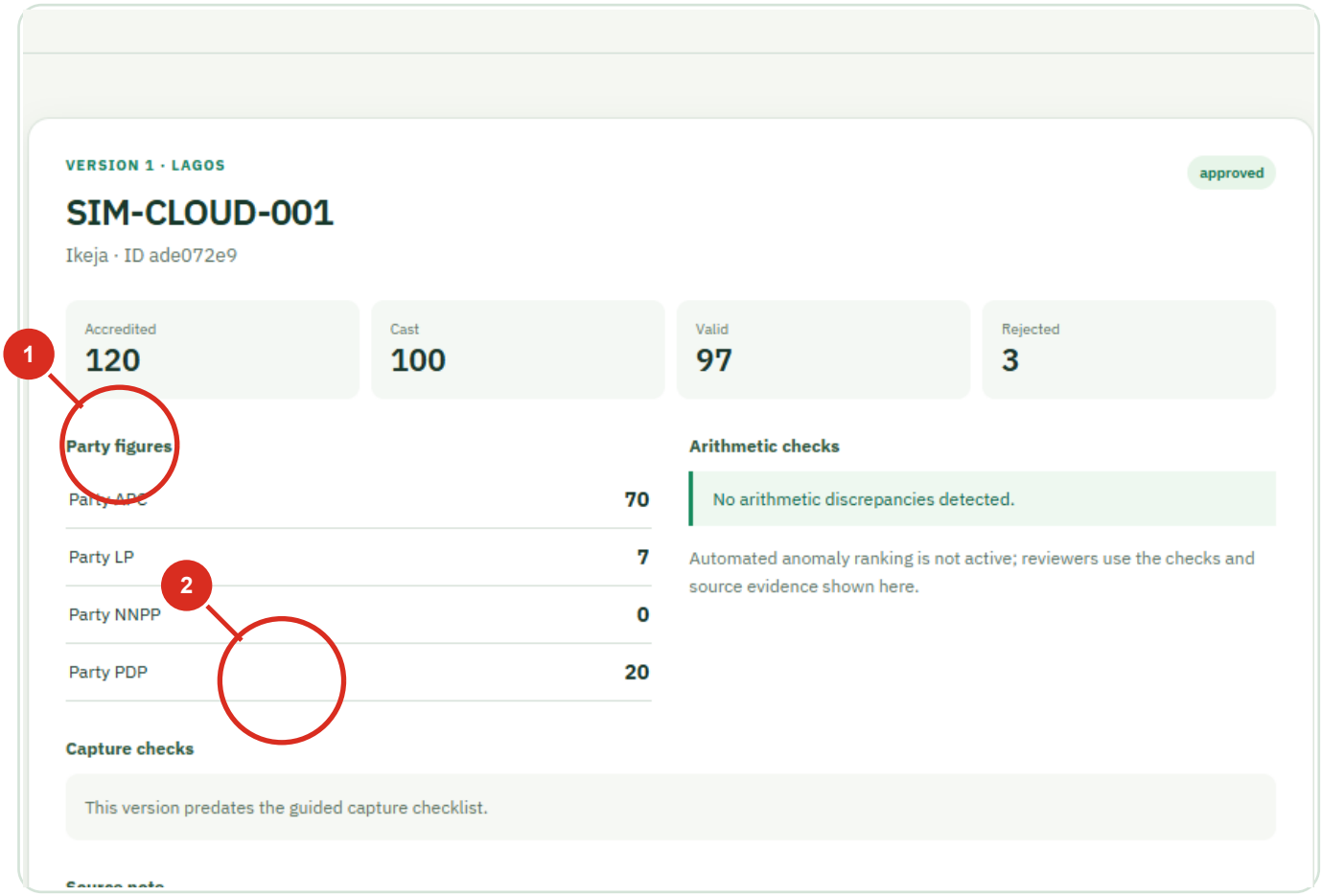

Ikeja LGA. Ward boundaries appear inside the LGA. Select the highlighted Aguda ward on the map or its row in the ward list.




Aguda selected. The regional panel names the ward and highlights its row. Use Received reports · Aguda below the list to open reports for that ward. The green shading shows submission coverage only.

# 3. Inspect the evidence

Open Evidence review and select a report. Compare the entered figures with the source copy.




### 1 Inspect the record

Check unit and location, accredited/cast/valid/rejected totals, every party score, source note and any arithmetic prompts. Investigate differences against the original copy.

### 2 Compare scores and checks

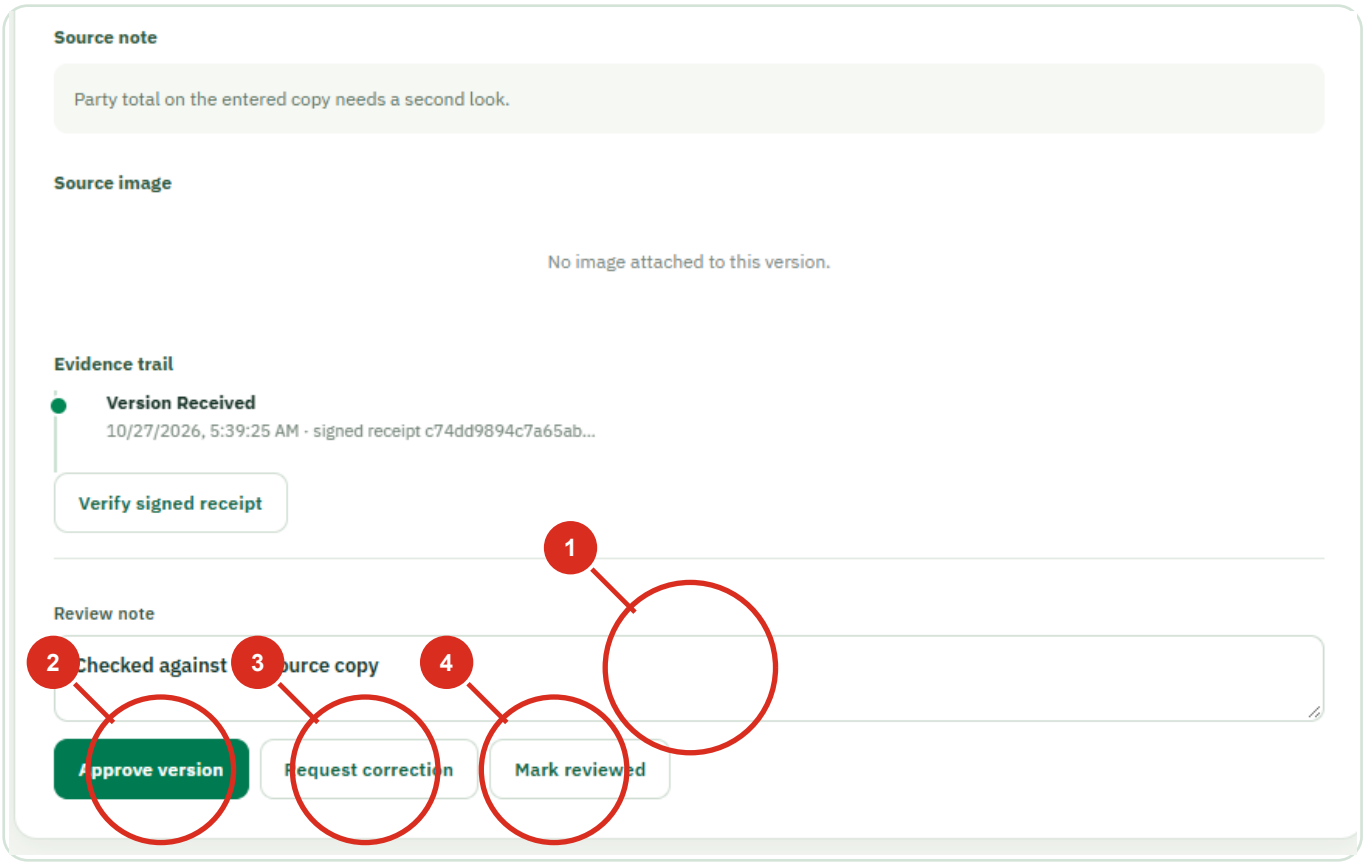
Inspect every party score and any arithmetic prompt against the source copy. A prompt calls for verification; it is not a conclusion.

Scroll farther down to inspect the source image, evidence trail and signed receipt before recording a decision. These controls are shown on the last page.

REVIEWER PATHWAY

# 4. Record a decision

The reviewer records a human decision for the selected version and can leave a clear note.




### 1 Write the review note

Summarize what you checked and what action the observer should take if a correction is needed. Do not treat a check alone as proof of wrongdoing.

### 2 Choose one action

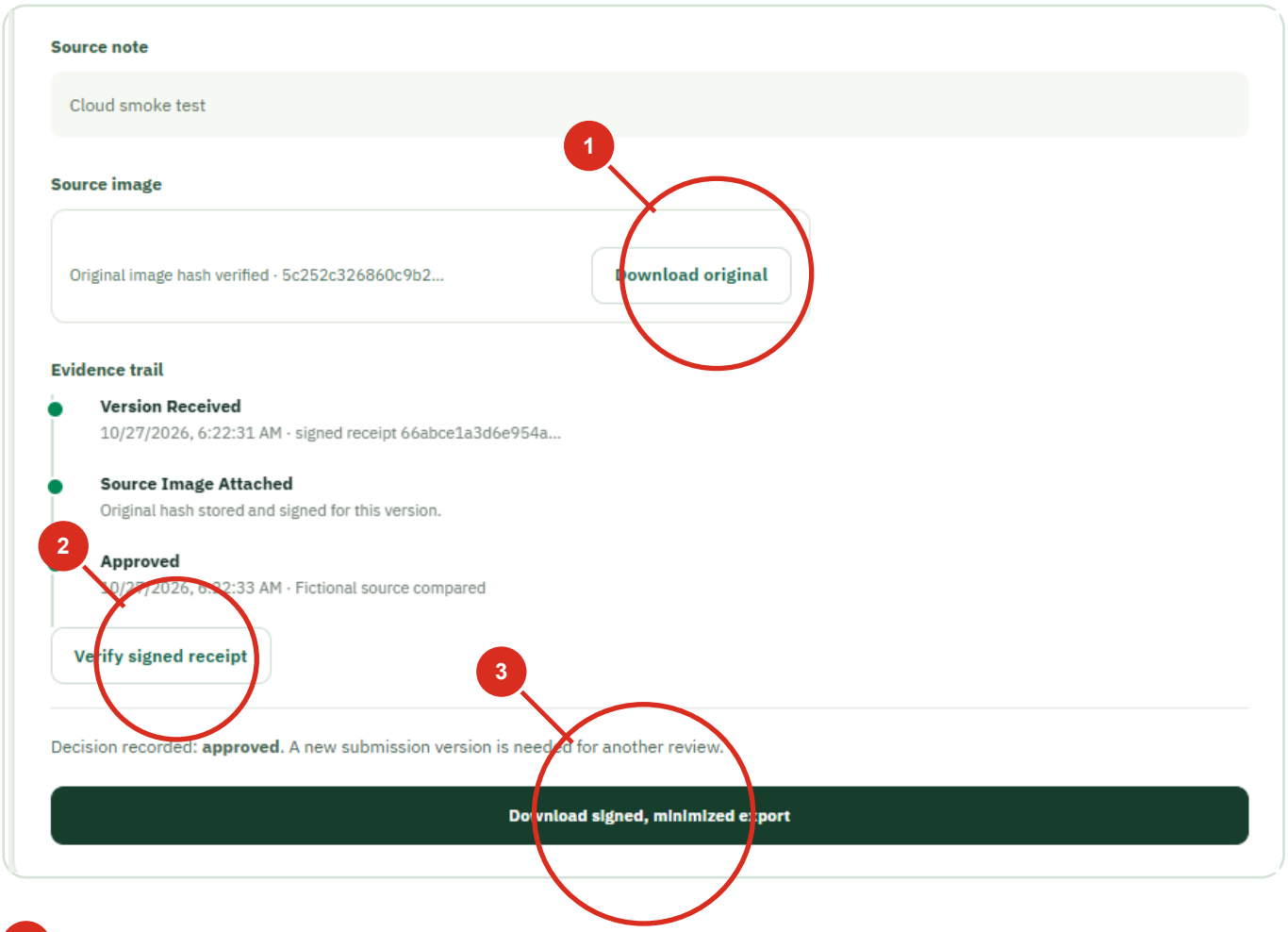
Use Approve version when the evidence supports it; Request correction when a new version is needed; or Mark reviewed to record review without approval.

For an approved version, the evidence view offers Download approved PDF. The observer sees the recorded decision and note in Report progress. A correction is submitted as a new version, preserving the earlier trail.

REVIEWER PATHWAY

# 5. Verify the evidence trail

Use the source image, receipt and export controls for the selected version.




### 1 Compare image and figures

If an image is attached, select Zoom image and inspect each entered figure. Record Matches image, Possible mismatch, Cannot read or Not compared with a reviewer note. Notes never change the submitted version.

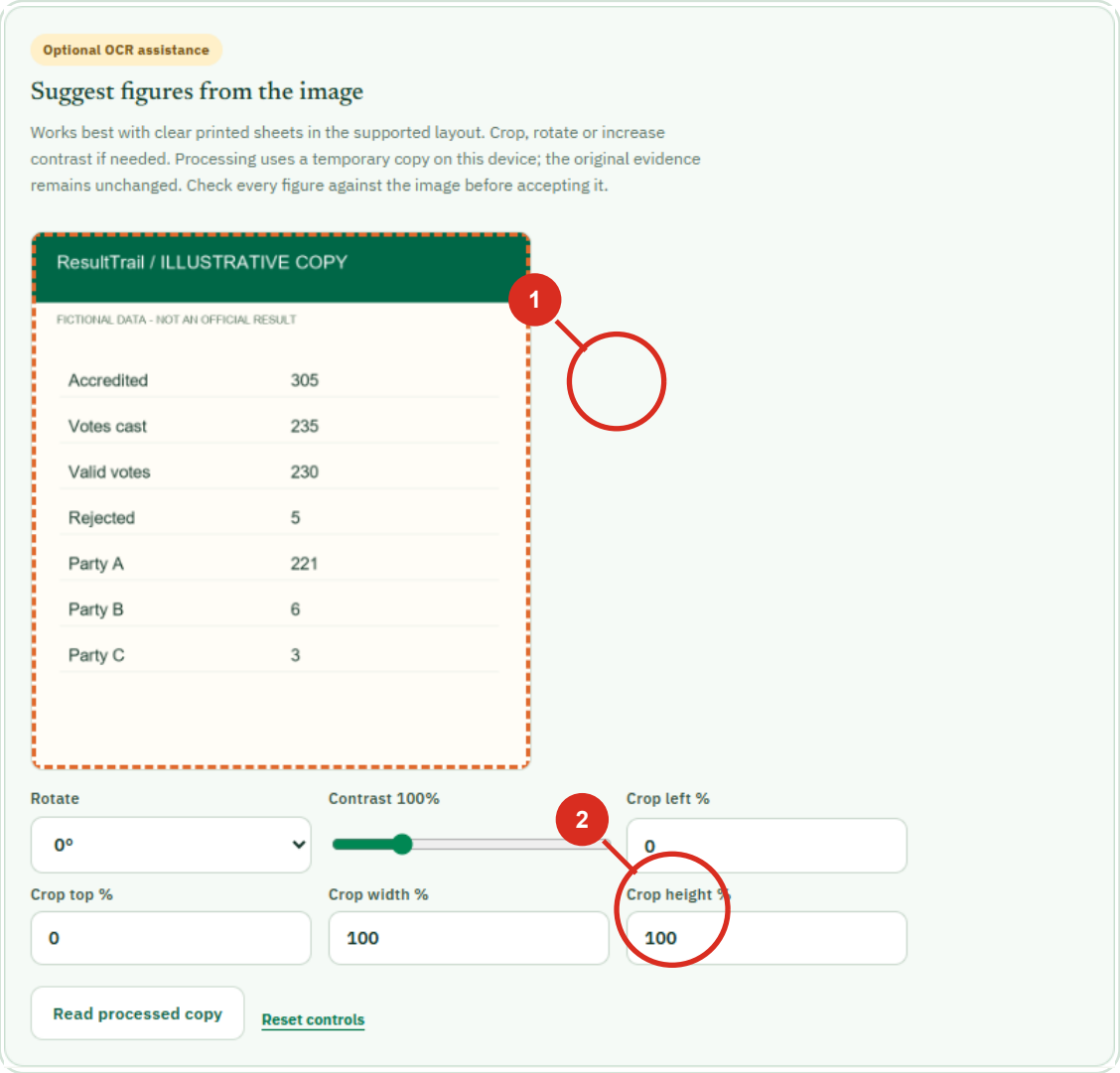
### 2 Verify the receipt

The comparison history preserves each reviewer action. The evidence trail shows when the version and image arrived. Use Verify signed receipt to check the signed report digest.

The comparison history keeps earlier human notes. Optional OCR hints appear below the comparison panel when a verified image is present. An approved version offers Download approved PDF.

# 5. Read an image with OCR

OCR is optional assistance for a clear printed fictional sheet. The original attachment stays unchanged.




### 1 Adjust a temporary copy

After attaching an image, use crop, rotation and contrast when needed. These controls affect only the copy sent to on-device OCR; the evidence image and hash stay unchanged.

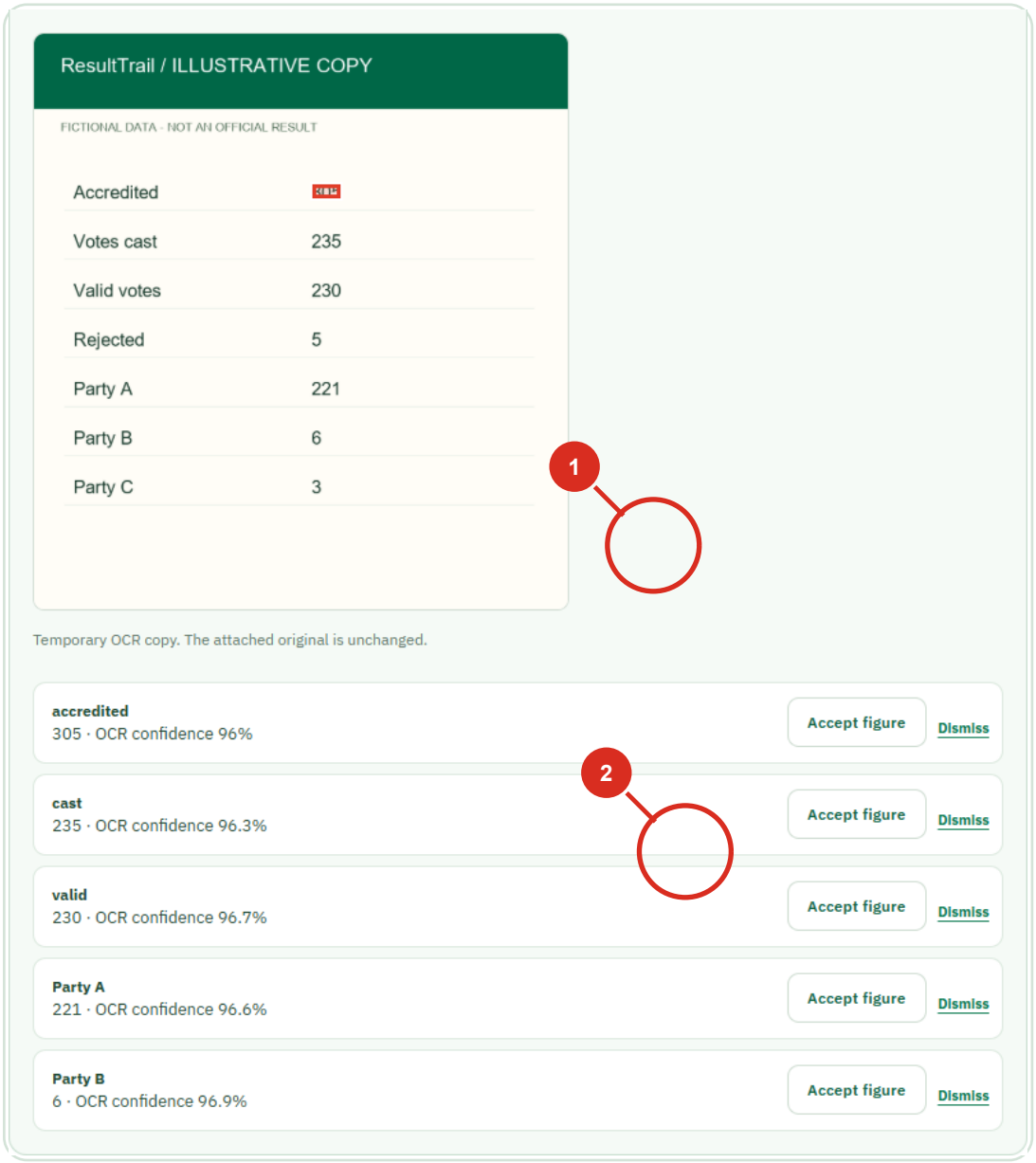
### 2 Read and inspect

Select Read processed copy. Processing runs in this browser. Check the processed image and the original before using any suggestion; OCR can misread or omit figures.

Supported printed layout matters. Handwriting, shadows, glare and skew can defeat the parser. If a figure is unclear, type it from the source instead of accepting a guess.

# 6. Accept figures one at a time

The OCR list is a set of suggestions, not a completed report.




### 1 Compare each proposed value

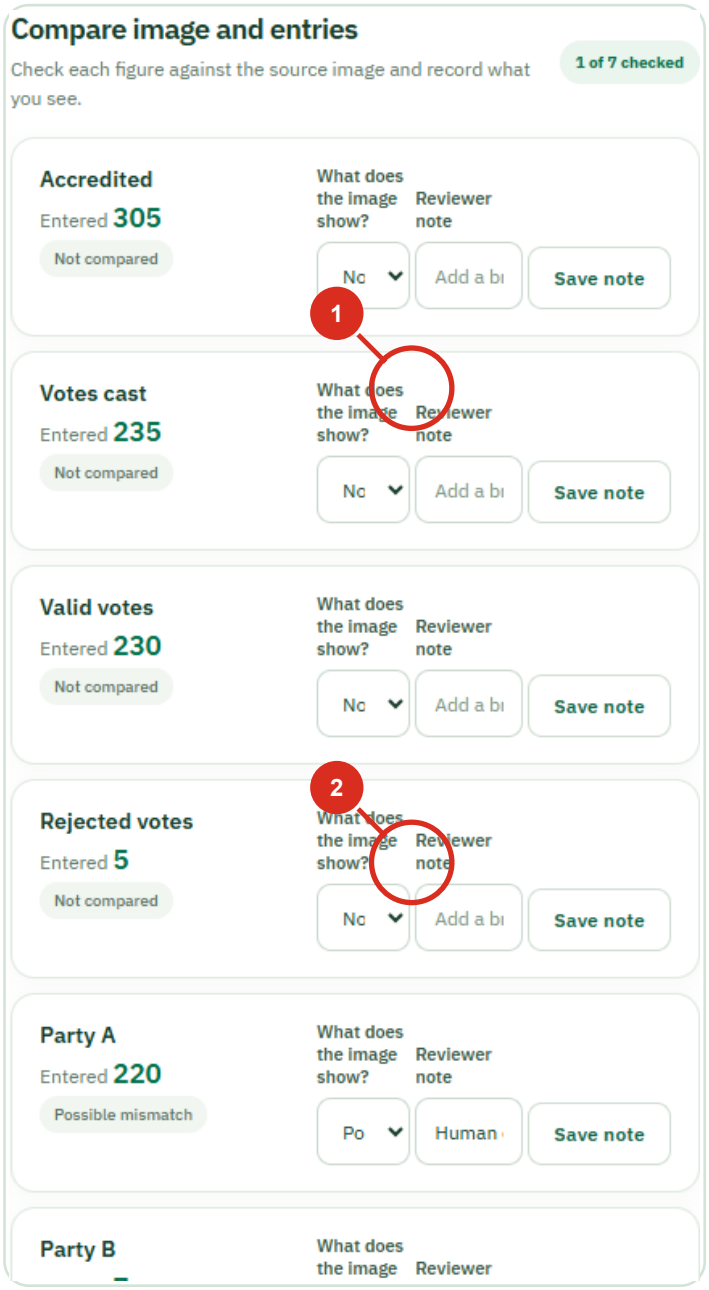
Select a suggestion to inspect its highlighted region. Compare the printed value with the suggestion and with the form field. The percentage is OCR confidence, not proof that the figure is correct.

### 2 Accept or dismiss

Select Accept figure only for a value you checked. Use Dismiss for an unreadable or wrong value. Every required field, party row and capture check still needs your review before saving.

# 6. Compare image and entries

For an attached image, record a human observation for each figure you inspect.




### 1 Choose the comparison

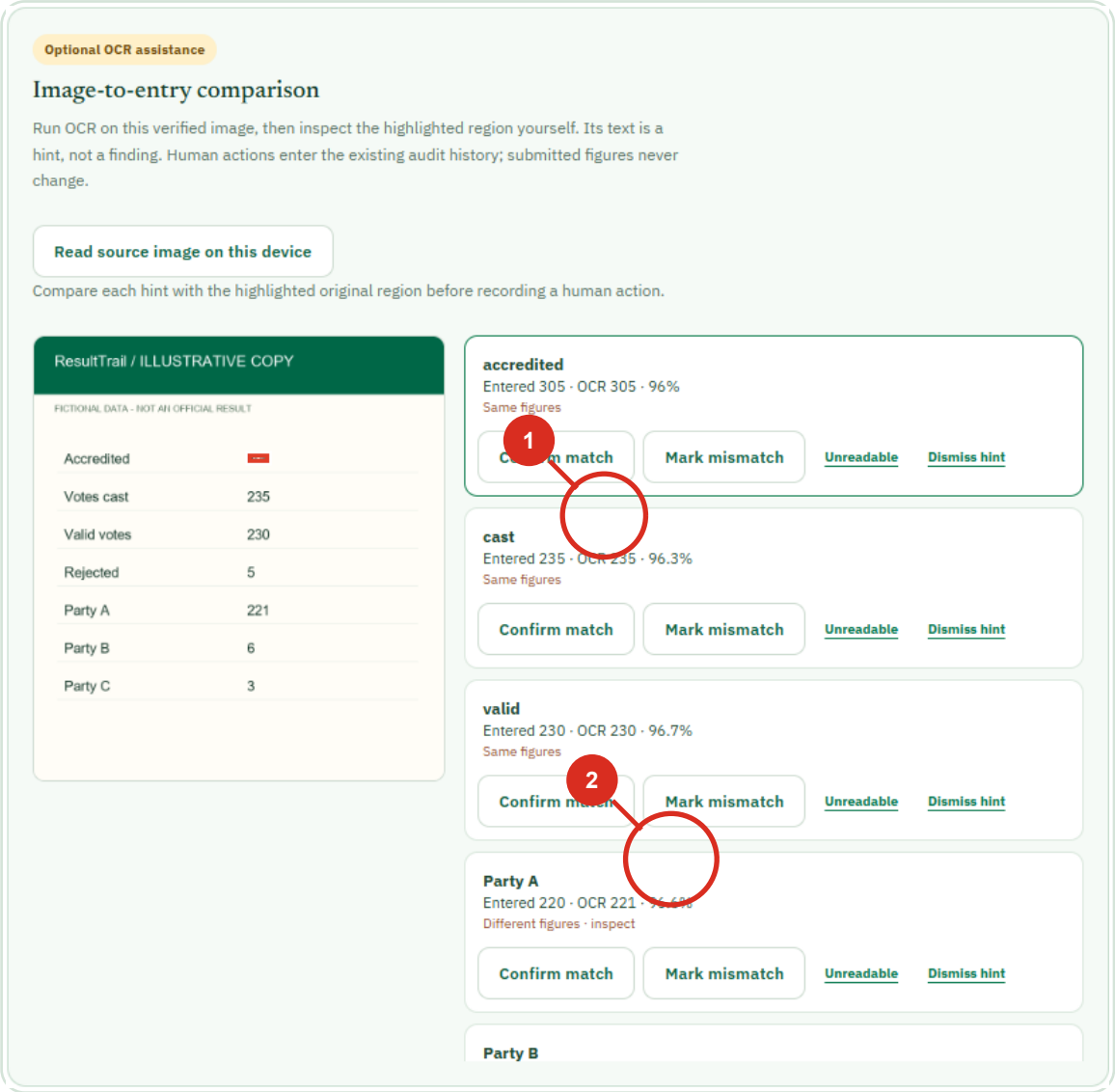
Use Matches image, Possible mismatch, Cannot read or Not compared. Add a brief note explaining what you saw, then save it.

### 2 Keep the trail intact

A saved comparison does not alter submitted figures. Later human notes can supersede an earlier assessment while the comparison history remains visible. Request a new reporter version when correction is needed.

# 7. Inspect optional OCR hints

OCR reads the verified image locally and highlights its proposed source region.




### 1 Run and compare

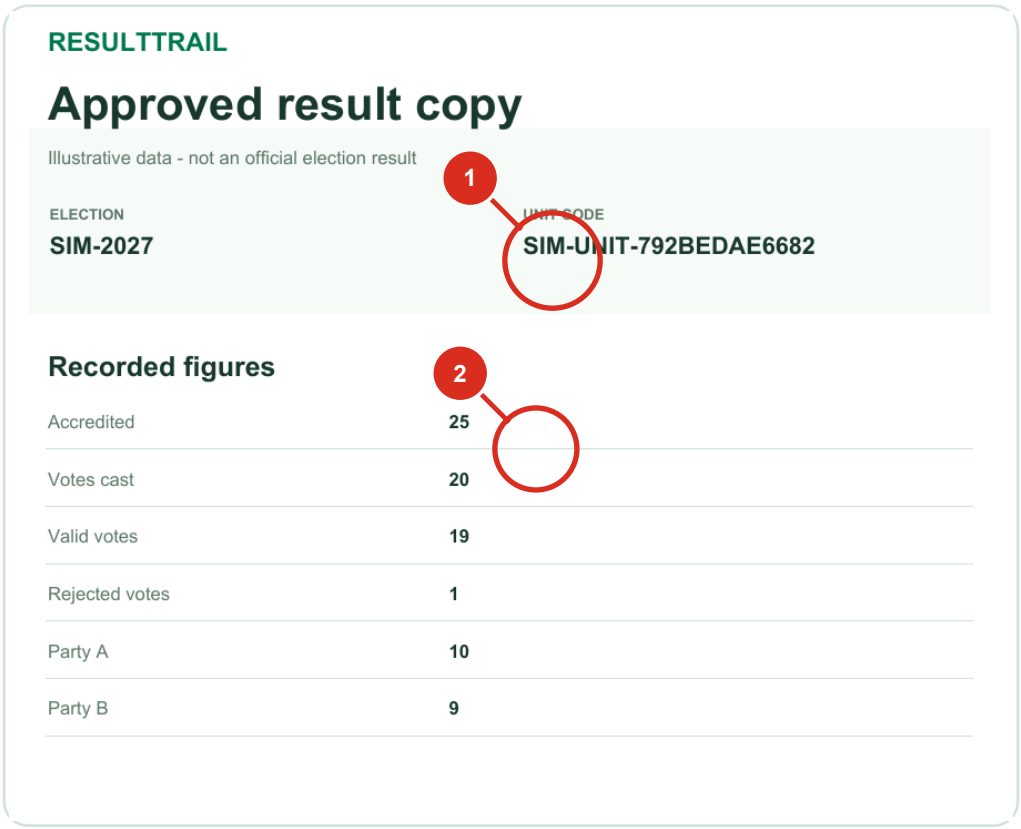
Select Read source image on this device. Select a hint to see its region on the original. Different figures call for inspection, not automatic rejection; equal figures still require visual confirmation.

### 2 Record the human action

Choose Confirm match, Mark mismatch, Unreadable or Dismiss hint. The action enters the existing comparison history. OCR never changes the received report or makes the approval decision.

# 8. Download the approved PDF

The PDF makes an approved illustrative result easier to read and share.




### 1 Approve the latest version

After checking the image and figures, record an approval. The selected report then shows Download approved PDF. Only an approved latest version can be exported.

### 2 Understand what is signed

The visible PDF pages summarize the approved figures. The exact signed, minimized JSON package and verification public key are embedded as PDF attachments. The PDF pages themselves are not digitally signed.

For formal verification, extract and verify the embedded signed package with the project verifier, then compare its figures with the visible PDF. A signed receipt proves server receipt, not authenticity of the photographed paper.

CREDITS AND SCOPE

# Map data attribution

Boundary outlines provide geographic context for the reviewer explorer.

State and LGA boundaries

The Nigeria state/FCT and LGA outlines shown in this guide are derived from geoBoundaries gbOpen Nigeria ADM1 and ADM2 simplified datasets, licensed CC BY 4.0. Source: geoboundaries.org.

Ward boundaries

The ward outlines are derived from GRID3 NGA - Operational Wards v4.0, copyright 2026 The Trustees of Columbia University in the City of New York, licensed CC BY-SA 4.0. Source: data.grid3.org. The simplified ward files used by ResultTrail are distributed under CC BY-SA 4.0.

How to interpret the map

GRID3 describes its ward boundaries as operational and not yet fully validated by government authorities. They may differ from official electoral wards or from the state/LGA layer. Green shading in ResultTrail counts illustrative reports received by this workspace; it does not show vote totals, election outcomes or a finding of irregularity.

For dataset details, source revisions and links, see docs/geodata.md in the ResultTrail repository.